CMPE- 257-01 PROJECT PROPOSAL

Early Prediction of Reading Performance in Young Learners

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1. Problem Description

In the initial days, the reading development is diverse across young students. Some students may show difficulties in word identification, can also have word attack, or oral reading before these problems become obvious through classroom performance. This project will use the assessment results and relevant student learning environment information to predict the reading performance and identify students who may require additional support in fututre.

Early prediction of the reading habit of the students can help educators respond before reading difficulties become more serious. A machine-learning model could analyze patterns in previous assessments and background features to estimate a student's future reading performance. These predictions could help teachers and reading specialists plan targeted instruction, monitor progress, and make better use of intervention resources. The model would be used as a decision-support tool and would not replace professional judgment or formal assessment.

The project may benefit students, classroom teachers, reading specialists, and school administrators. It focuses on early-childhood and elementary reading education, using student performance data to support timely and individualized educational interventions.

2. Dataset

Research paper:
https://papers.neurips.cc/paper_files/paper/2024/hash/862170d83a17b7c2ef3c5a58e7f1992d-Abstract-Datasets_and_Benchmarks_Track.html

Dataset: https://github.com/ecri-data/ECRI-Dataset/tree/main/ECRI

The Enhanced Core Reading Instruction(ECRI) dataset is a large longitudinal tabular dataset that is designed for predicting the future reading performance of students in early childhood education. This data is collected from 44 distinct schools with 6,916 students and 172 teachers. This dataset is a collection of information that tells us who the students are, what are their scores under different tests and also what sought of extra help is being offered by the school to the students to overcome their reading difficulties. We have various parameters including special education status, gender,age, different test scores and also number of at-risk readers in class. In short, they are a combination of demographic, assessment and intervention data.

As missing data is something that is common with ECRI dataset, we have the need for preprocessing the data. There are different steps in it which includes replacing missing values with some meaningful aggregated data, converting text in columns like gender and groups to numbers using one hot encoding. Because different tests have different scoring scales, we need to convert them into a standardized test score so that no specific parameter can overperform the others. We also find anomalies that might affect the data distribution and correct them. Once all these steps are done, we start feature selection in order to ensure only the data before the required prediction period is used for training and the data is split into three different groups like one for training, another for

validation and the rest for testing. We also inspect the data both at school level and also at student level so that we can train the model only on a certain set of data and see how the model can generalize on new unseen data.

3. Initial Approach

This project focuses on predicting the future reading outcomes based on the students past available information. We don't always have complete data available for all students so in order for the model to deal with such scenarios we use MaskMLP which hides a certain set of data during training so that the model can make accurate predictions even when partial data is missing. We first focus on understanding the relationship between the data and train different basic ML models like Linear Regression or Decision tree to create a reference point. We then use ensemble models like Random Forest, XGBoost and a neural network (Multi Layer Perceptron) to understand the sophisticated relationship between the target variable and the rest of the variables. We then use the maskMLP approach and compare the prediction of all the models. We plan to evaluate the model under different settings, one with the school level set and the other with student level. We then evaluate the model's performance based on metrics like accuracy and Area Under the Curve as the prediction is going to be whether a student can show good reading progress or not.

4. Expected Challenges

The main challenge is on how we handle missing data as all students would not have complete details available and it is essential to decide on where to use MaskedMLP approach and where to fill aggregated values. The next challenge is class imbalance where one reading progress category has more number of students than the others making the category to outperform and create inaccurate predictions. Data leakage is an important factor we need to focus on in order for the model to only use past data during training and we should finetune the model so that it doesn't overfit.

5. Team Roles

All team members(Anupama, Megha, and Supriya) will collaborate on basic data preprocessing, documentation, report writing, presentation preparation, and final review of the project.

Team Member	Responsibility
Anupama Singh	Baseline model training and evaluation - Implement Linear Regression and Decision tree regressor to set up baseline performance and evaluate using error metrics
Megha Gangal	Ensemble models - Implement Random Forest and XGBoost models and conduct feature-importance analysis, and interpret factors affecting model performance.
Supriya Kavitha Venkatesan	Neural Network and Comparison analysis - Implement MLP/MaskMLP-inspired model, visualize model performance and compare the traditional ml and neural network approaches.

6. References

Shangguan, Z., Huang, Z., Ohn-Bar, E., Ozernov-Palchik, O., Kosty, D., Stoolmiller, M., & Fien, H. (2024). Scalable Early Childhood Reading Performance Prediction.

Enhanced Core Reading Instruction dataset and project documentation. https://ecri-data.github.io/